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An Efficient Hybrid Deep Learning Framework for CO₂ Plume Monitoring in Geological Sequestration

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Abstract

Geological CO₂ sequestration (GCS) represents a critical technology for mitigating CO₂ emissions, but faces significant challenges in environmental monitoring and risk assessment. While physics-based reservoir simulations provide accurate predictions, their computational cost severely limit practical applications. Deep learning-based surrogate models offer 100 to 1000 times speedup but meet training difficulties for large-scale systems. Moreover, practical GCS monitoring relies on seismic technologies limited to detecting CO₂ saturation changes, necessitating rapid predictive methods aligned with real-world detection capabilities. This study introduces a novel deep learning workflow for GCS to predict seismicity-detectable CO₂ plume fronts. Following established practices for seismic monitoring data interpretation, continuous saturation plume fields are converted to binary maps using a 10% CO₂ saturation threshold, to mimic the practical detection limit of current seismic technologies. The workflow employs a hybrid CNN-Transformer architecture. Convolutional neural networks (CNNs) excel at spatial feature extraction, while transformer networks have strength in sequence modeling. In order to capture complex spatiotemporal CO₂ plume evolution patterns, the framework comprises: a geological encoder using 3-dimensional (3D) CNNs to process high-dimensional geological properties (porosity and permeability), a well-control encoder coupled with Transformer blocks to handle dynamic well control sequences, and ultimately a decoder to reconstruct seismicity-detectable saturation distribution. Based on the large-scale 3D SPE11-C reservoir model, the hybrid CNN-Transformer model reaches an average binary cross-entropy (BCE) loss at 0.099 ± 0.0788 and Dice score at 0.799 ± 0.181 , and achieves nearly 10,000 times (on GPU) and 3,500 times (on CPU) acceleration over reservoir simulation in the prediction. Therefore, the hybrid CNN-Transformer model successfully captures essential spatiotemporal dynamics of CO₂ plume evolution, and is expected to provide a better balance between computational speed and physics fidelity.

Keywords: Geological CO₂ Sequestration, Deep Learning, CNN-Transformer, CO₂ Plume Monitoring, Reservoir Heterogeneity

Introduction

Geological CO₂ sequestration (GCS) is a critical technology for achieving global net-zero emissions, with deployment accelerating rapidly to meet climate targets (Hoang et al., 2022; IPCC, 2023; Zhi et al., 2023). However, large-scale implementation faces critical challenges in environmental monitoring and risk assessment (Alcalde et al., 2018; Harp et al., 2016; Miocic et al., 2019), particularly in detecting CO₂ plume migration with sufficient sensitivity. Current seismic technologies for practical CO₂ plume monitoring can only resolve saturation changes above 5–10% thresholds, which limits the fidelity of plume characterization (Roach et al., 2017). While physics-based reservoir simulations provide high-fidelity predictions of multiphase flow dynamics, their prohibitively high computational cost severely restricts applications in ensemble-based risk assessment, real-time monitoring, and operational optimization (Feng et al., 2025; M. Tang et al., 2021; Tariq et al., 2023; Yan et al., 2022a). Since seismic monitoring remains the primary practical method for CO₂ plume detection at field scales, surrogate models must operate within these observable detection limits to ensure their predictions can be validated against real monitoring data and inform actionable risk assessment decisions. This creates an urgent need for computationally efficient surrogate models that directly align with seismicity-detectable saturation thresholds and preserve essential spatiotemporal accuracy for large-scale GCS sites.

Deep learning surrogate models have emerged as a promising solution, achieving computational speedups of 100 to 1000 times over traditional simulators while maintaining reasonable accuracy for reservoir forecasting (Fan et al., 2025; Meng et al., 2024; Xu et al., 2023). Advanced architectures such as Fourier Neural Operators (FNOs) (Tang et al., 2024; Wen et al., 2023, 2022; Yan et al., 2022a), Convolutional Neural Networks (CNNs) (Mo et al., 2019; Tang et al., 2021a; Yan et al., 2022b, 2022c; Zhong et al., 2019), and Transformer (Feng et al., 2025; Liu et al., 2025) have demonstrated efficacy in subsurface flow modeling. However, these approaches face persistent challenges when applied to practical GCS monitoring scenarios. Significant training cost and memory consumption hinder their deployment in large-scale 3D GCS models. Moreover, most of existing methods generate continuous saturation outputs that disregard the fundamental limitations of field seismic monitoring, where only saturation changes exceeding a threshold of 5–10% are practically detectable. Tang et al. (2021b) resolved this mismatch by projecting 3D saturation fields predicted by deep learning models into 2D binary maps using a 10% threshold to represent seismicity-detectable plume fronts. However, this approach is not yet end-to-end. This dual challenge of computational scalability and seismic relevance present a pressing need of efficient and accurate predictive models for CO₂ plume front tracking during practical operational implementation.

This work introduces a novel a hybrid CNN-Transformer model for GCS surrogate modeling. It processes heterogeneous 3D reservoir properties (e.g., porosity and permeability) via 3D CNN to capture spatial complexity, while dynamic well controls and scalar reservoir parameters are encoded through multilayer perceptron (MLP). Further, the latent representations from both pathways are fused and processed through transformer blocks (Vaswani et al., 2017), leveraging the self-attention mechanism to identify critical dependencies between geological features and operational parameters across temporal sequences. This integration enables the model to dynamically weight influential factors during CO₂ plume evolution. The framework uniquely aligns with seismic monitoring by converting outputs to binary saturation maps using a threshold of 10% for CO₂ saturation, representing seismicity-detectable plume fronts. Validation using the 11th Society of Petroleum Engineers Comparative Solution Project (SPE11-C) reservoir model (Nordbotten et al., 2024) demonstrates the architecture's capability to accurately predict CO₂ plume fronts, while achieving computational speedups of nearly four orders of magnitude as compared to traditional reservoir simulation.

Methodology

The methodology of this work mainly includes three parts: model formulation, CNN-Transformer architecture description, and simulation data generation.

Model Formulation

In this study, the GCS process is typically two-phase (supercritical CO₂ and brine water phases) flow in porous media, which is governed by two-phase mass conservation. The two-phase mass conservation equation can be written as the following,

$$\frac{\partial}{\partial t} \left(\phi \sum_{\beta} S_{\beta} \rho_{\beta} x_{\beta}^i \right) - \nabla \cdot \left(K \sum_{\beta} \frac{k_{r,\beta} \rho_{\beta} x_{\beta}^i}{\mu_{\beta}} (\nabla p_{\beta} - \rho_{\beta} \mathbf{g}) \right) - \sum_{\beta} \rho_{\beta} x_{\beta}^i q_{\beta} = 0 \quad (1)$$

where the first term is the fluid accumulation within the porous media, the second term on the left side is the mass flux governed by extended multiphase Darcy's law, and the third term is the source or sink term. The subscript β illustrates fluid phase (CO₂ or brine); ϕ is the porosity of the porous media; K is the absolute permeability; x_{β}^i is the mole fraction of component i (CO₂ or H₂O) in phase β ; $k_{r,\beta}$ is the relative permeability; S_{β} is the phase saturation; ρ_{β} is the phase density; p_{β} is the phase pressure; μ_{β} is the phase viscosity; $\mathbf{g}$ is the gravitational acceleration; q_{β} is the well volumetric flow rate.

The phase pressures p_{β} is related through the capillary pressure p_c , shown as below,

$$p_c(S_{\text{CO}_2}) = p_{\text{CO}_2} - p_{\text{Brine}} \quad (2)$$

In typical reservoir simulation, after discretizing Eq. (1), state variables like pressure and saturation can be calculated iteratively by numerical methods, which are high-dimensional variables across time and space scales. Let $\vec{m}$ represent high-dimensional inputs of reservoir simulation, e.g., permeability and porosity fields, and $\vec{y}$ be the corresponding high-dimensional state variables, e.g., saturation and pressure, then the forward reservoir simulation operator $f(\cdot)$ can be abstracted as,

$$\vec{y} = f(\vec{m}) \quad (3)$$

Deep learning-based surrogate model with specifically designed workflow can efficiently capture the nonlinear mapping relationship $f(\cdot)$ embedded in high-dimensional input and output variables, and ultimately achieves similar accuracy to numerical simulators but with significant speedup, shown as below,

$$\vec{y} \cong \text{NN}(\vec{m}; \theta) \quad (4)$$

where $\text{NN}(\cdot)$ represents the deep learning-based surrogate model; θ indicates the learnable parameters within the surrogate model.

Since this study focuses on predicting seismicity-detectable CO₂ plume fronts, $\vec{y}$ specifically denotes snapshots of binarized S_{CO_2} , which requires specific saturation level as the threshold δ . Inspired by Tang et al.'s work (2021b), $\delta = 0.1$ is used to binarize the original saturation fields, as follows:

$$S_{\text{CO}_2}^b = \begin{cases} 0, & S_{\text{CO}_2} < \delta \\ 1, & S_{\text{CO}_2} \geq \delta \end{cases} \quad (5)$$

where S_{CO_2} is the original saturation value; $S_{\text{CO}_2}^b$ is the binarized S_{CO_2} , whose value is 1 for CO₂ plume and 0 for non-CO₂ plume.

Although the binary threshold approach discards continuous saturation information, it retains the essential spatial patterns detectable through seismic monitoring, realistically capturing critical plume migration features and fault-induced discontinuities of fluid distributions. Fig. 1 exhibits the information retention capability of this binary threshold approach by comparing original and processed saturation fields. Using

a representative 2D slice from the test dataset, Fig. 1(a) demonstrates how continuous saturation gradients are simplified, while Fig. 1(b) and (c) crucially preserve critical geological features, particularly the distinct fault-induced discontinuities that remain clearly identifiable in the binarized output. These structurally significant patterns provide actionable insights for plume management and risk assessment.

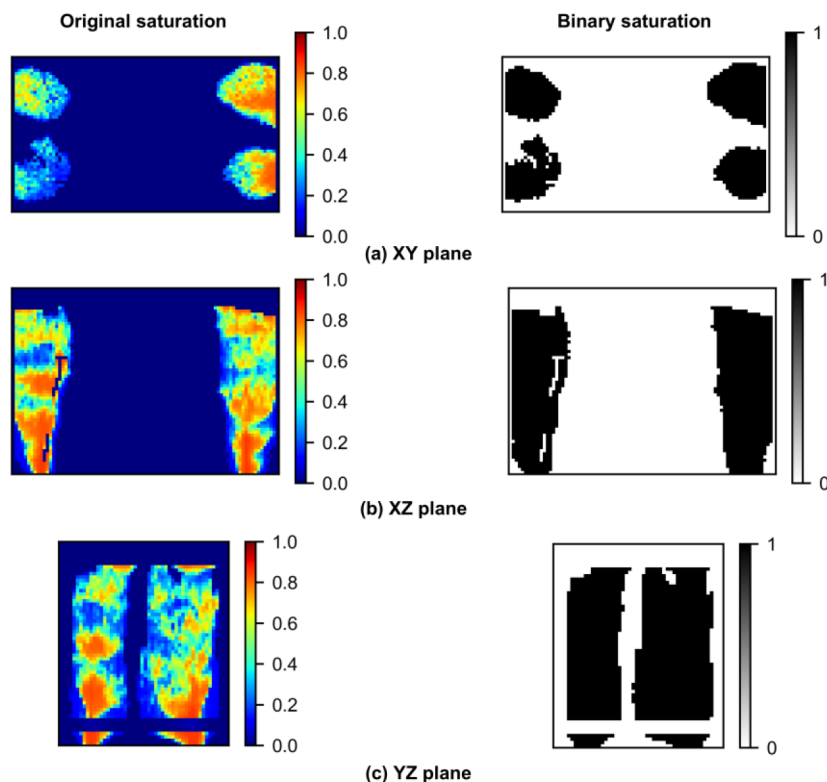


Figure 1—Schematic diagram of the binarization process using 0.10 as the CO₂ saturation threshold. The 30th time step from testing Case 88 is taken as the example). (a) 10th X-Y plane; (b) 10th X-Z plane; (c) 10th Y-Z plane.

CNN-Transformer architecture

Conventional CNN-based surrogate models process high-dimensional geological features (permeability, porosity fields) alongside low-dimensional operational parameters through unified architectures. It requires manually broadcasting dimensions of common scalar reservoir input variables (e.g., reservoir temperature) to accommodate the mandatory data structure in CNN. This approach usually works but also incurs substantial memory consumption. On the other hand, the hybrid CNN-Transformer architecture (Feng et al., 2025) overcomes this limitation through specialized pathway processing. It takes 3D convolutional networks to extract high-dimensional spatial features directly from reservoir properties while preserving structural integrity, and MLPs to efficiently encode dynamic well controls and scalar parameters. In this work, we continue using this architecture and examine its capability to predict CO₂ plume front in a realistic large-scale GCS site.

As illustrated in Fig. 2, the hybrid CNN-Transformer architecture comprises several distinct components: a high-dimensional data encoder processing geological properties (Fig. 2(a)), a low-dimensional data encoder handling operational parameters (Fig. 2(b)), Transformer blocks modeling feature interactions (Fig. 2(c)), and a final reconstruction decoder (Fig. 2(d)). Crucially, 3D reservoir properties are compressed into latent representations that is combined with MLP-processed operational variables in a fixed-dimensional space (Fig. 2(f)), avoiding dimensional expansion, while enabling efficient processing through Transformer blocks. Here, the self-attention mechanisms dynamically capture spatiotemporal dependencies between

geological features and operational sequences, enabling accurate saturation prediction with less memory consumption.

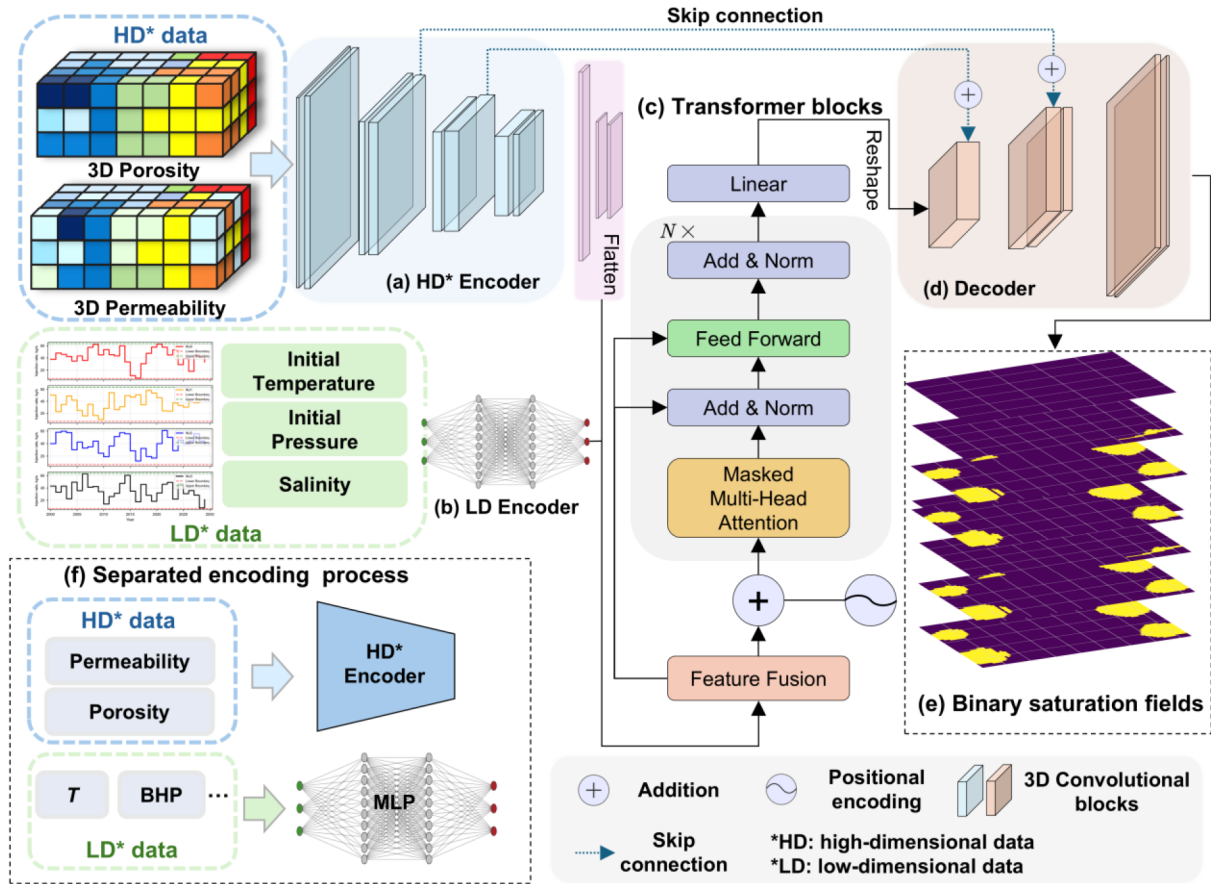


Figure 2—CNN-Transformer architecture (a) High-dimensional data encoder; (b) Decoder (reconstruct high-dimensional data); (c) Transformer blocks; (d) Output binary saturation fields; (e) Separated encoding process illustration.

Furthermore, our framework generates seismicity-detectable outputs by applying a 10% saturation threshold to create binary plume segmentation maps. This approach provides direct alignment with field monitoring capabilities since seismic technologies detect only saturation changes exceeding this threshold. By converting continuous regression to discrete segmentation, the prediction stability is enhanced through exclusive focus on detectable plume fronts rather than unresolvable saturation gradients. Therefore, the outputs explicitly delineate fault-impacted migration pathways of CO₂ as discrete spatial features, enabling immediate risk assessment of CO₂ plume containment.

The binary cross-entropy (BCE) measures how much the difference is between the model predictions and the actual binary labels. As a result, the BCE loss is employed as the primary training loss function for the segmentation task. It can optimize pixel-wise classification accuracy through directly penalizing deviations between predicted plume boundaries and seismicity-detectable regions (ground truth). The BCE loss is defined as,

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N \left[y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i) \right] \quad (6)$$

where y_i denotes the true S_{CO_2} (0 or 1) at voxel i ; $\hat{y}_i$ represents the predicted probability of CO₂ presence; N is the total number of spatial units. Usually, a perfect prediction would give BCE = 0, while a random poor prediction typically gives BCE $\approx$ 0.693.

Additionally, the Dice coefficient emphasizes spatial overlap and boundary accuracy over individual pixel classification probabilities. Dice is especially suitable for GCS monitoring where thin and discontinuous plumes due to faults demand precise boundary delineation rather than bulk saturation accuracy. Therefore, the Dice coefficient is treated as a key evaluation metric to quantify spatial alignment of predicted plume geometries, which can help assess fault-impacted migration pathways. This Dice coefficient is calculated through,

$$\text{Dice} = \frac{2 \cdot |Y \cap \hat{Y}|}{|Y| + |\hat{Y}|} \quad (7)$$

where Y and $\hat{Y}$ denote ground-truth and predicted binary spatial units; $|Y|$ is the total number of pixels in spatial unit Y ; $|\hat{Y}|$ is the total number of pixels in spatial unit $\hat{Y}$; $|Y \cap \hat{Y}|$ is the number of pixels in both spatial.

Simulation Data Generation

Several key modifications are implemented on the 3D SPE11-C benchmark model to enhance its applicability for GCS projects. Specifically, the reservoir thickness is reduced from 1200m to 300m to better represent typical GCS target formations. Further, we relocate well placement for dynamic well control scenarios, and incorporate more realistic heterogeneous distributions of porosity and permeability fields. The resulting modified saline aquifer model features complex geometries with heterogeneous geological properties (Fig. 3(a) and (b)) and multiple faults (Fig. 3(c)). The spatial-temporal evolution of CO₂ saturation during injection stage is simulated using the commercial reservoir simulator CMG-GEM (CMG, 2022). The reservoir domain utilizes a high grid resolution comprising $86 \times 50 \times 60$ cells (in total 258,000 cells), in order to capture fine-scale geological heterogeneities and CO₂ migration patterns.

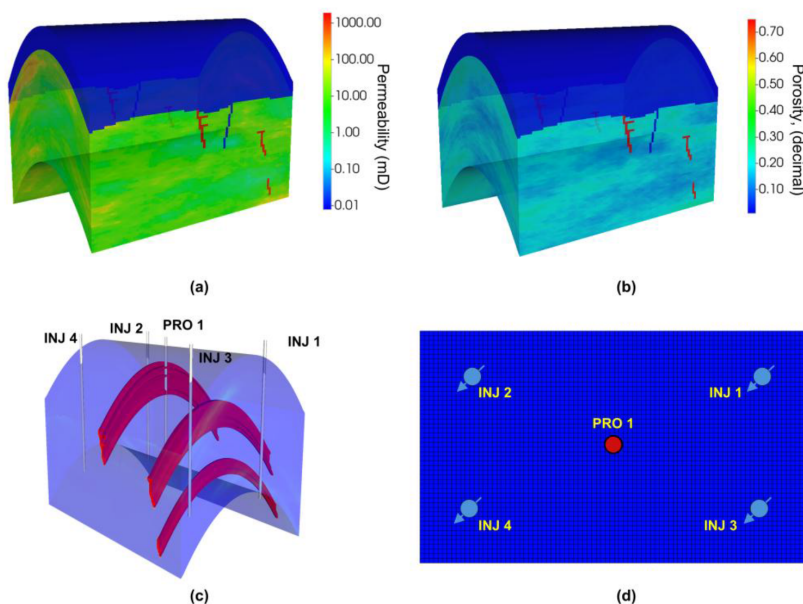


Figure 3—Modified SPE11-C simulation model. (a) Top view of well placement; (b) Visualization of multiple faults (in red color); (c) Heterogeneous permeability field; (d) Heterogeneous porosity.

The model incorporates realistic geological properties, including spatially varying permeability (ranging from 0.5 to 560 mD), porosity (0.07 to 0.27). Injection wells are in the four corners of the formation (Fig. 3(d)), with injection rates ranging from 0.2 to 2 million tons/year. The wells are perforated from layer 50 to layer 60. The only production well is in the center of the formation (Fig. 3(d)), with bottom hole pressure (BHP) ranging from 1 MPa to 5 MPa. No-flow boundary conditions are imposed at reservoir margins, with initial conditions set to hydrostatic pressure distribution and brine-saturated aquifer conditions. The

coupled flow-transport equations for CO₂-brine systems become nonlinear by incorporating the effects of compressibility, capillary pressure, and relative permeability. Table 1 comprehensively summarizes reservoir model parameters.

Table 1—Summary of reservoir model parameters

Parameters	Value/Range	Unit
Simulation Time	30	year
Number of Grids	86×60×50=258,000	/
Reservoir Size	8,400×5,000×300	m
Injection well number	4	/
Production well number	1	/
Permeability	0.5 to 560	mD
Porosity	0.07 to 0.27	/
Salinity	10,000 to 99,000	ppm
Reservoir temperature	60 to 100	°C
Reservoir pressure	13230 to 14850	KPa

Fig. 4 shows four examples of dynamic well controls, enforcing the injection rate and production BHP follow the fixed ranges and the difference between nearby time steps does not exceed one quarter of the whole range. To generate a comprehensive training dataset, we employ Latin Hypercube Sampling (LHS) to create 1,148 simulation cases in total. Each case has unique heterogeneous porosity fields and permeability with spatially correlated distributions, and distinct dynamic well control sequences of injection rates (Fig. 4(a)) and BHP (Fig. 4(b)). Reservoir conditions (temperature, salinity, initial pressure) vary to represent diverse storage scenarios. Simulations are performed to simulate 30-year injection of CO₂ into the model, and annually output 3D CO₂ saturation fields. The dataset is partitioned into training (918 cases), validation (115 cases), and blind testing (116 cases) sets to ensure rigorous model training and evaluation.

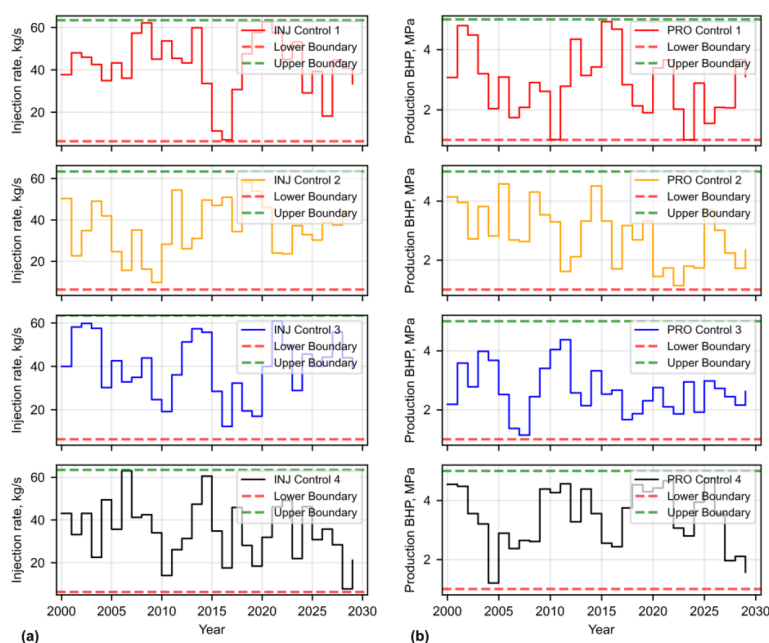


Figure 4—Dynamic well controls. (a) Four different dynamic injection rate schedules; (b) Four different dynamic production BHP schedules.

Results

The CNN-Transformer is trained using the AdamW optimizer (Loshchilov and Hutter, 2019). During training, we set the weight decay at 0.05, the learning rate at 0.001, and a batch size of 8. The maximum number of training epochs is 800, and early stopping is introduced to automatically stop the training process when the validation loss no longer improves. We trained and validated the surrogate model on an NVIDIA RTX A6000 GPU. Fig. 5 shows the evolution of BCE loss during the model training phase, including training loss and validation loss. The training stage automatically stops after 364 epochs due to the convergence of validation loss. The minimum training and validation BCE losses are 0.0437 and 0.0944, respectively. Because of the large weight decay setting and similar reduction trend of both losses, there are no apparent over-fitting issues. Furthermore, a low level of BCE loss indicates that our surrogate model can effectively capture the spatial-temporal evolution of CO₂ plume fronts.

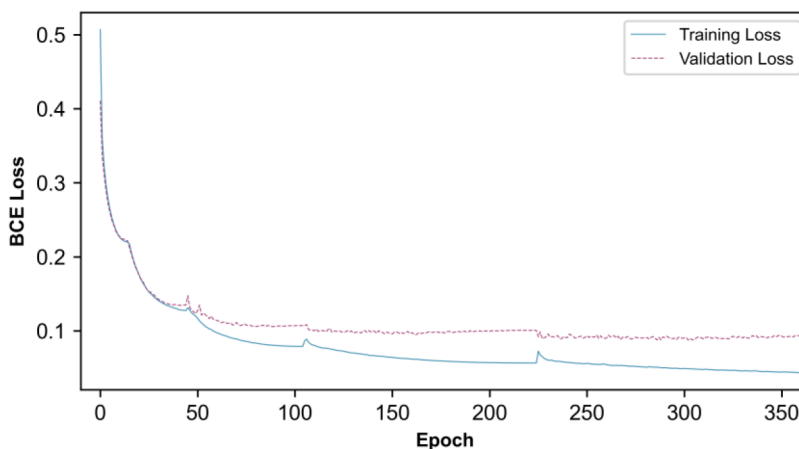


Figure 5—Training and validation BCE loss vs. epoch.

Fig. 6 presents a comparative analysis of CO₂ plume evolution predictions versus ground-truth simulations for a representative testing case (Sample 13) across four monitoring time steps: 10, 15, 20, and 30 years. Fig. 6(a) shows binary saturation predictions of the CNN-Transformer, where red voxels indicate seismicity-detectable CO₂ plumes (saturation >10%). Fig. 6(b) shows the corresponding high-fidelity simulation results serving as validation benchmarks. Fig. 6(c) shows the differences between Fig. 6(a) and (b), in order to quantify the spatial prediction accuracy. Through comparison, the Dice coefficients maintain remarkably stable, consistently exceeding 0.84 throughout the 30-year injection period, demonstrating the model's robust capacity for preserving plume morphology and boundary coherence despite increasing geometric complexity. Conversely, BCE loss exhibits a systematic increasing trend (from 0.059 to 0.153), reflecting the cumulative impact of spatially expanding CO₂ plume under continuous CO₂ injection. This progressive increase correlates directly with plume growth, where even minor pixel-level classification uncertainties compound across increasingly larger spatial extents. The observed metric behavior thus validates our loss function's sensitivity to operational reality: as injection continues, minor boundary inconsistency accumulates across larger plume surfaces, increasing BCE while Dice remains stable due to preserved shape coherence.

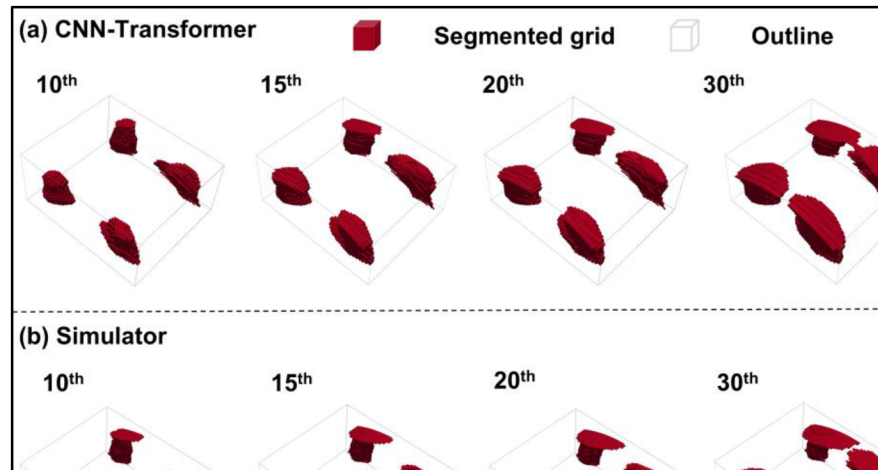


Figure 6—Comparison of $s_{\text{CO}_2}^b$ based on a representative testing case (13th sample) at four time steps. (a) Prediction by the CNN-Transformer; (b) Prediction by reservoir simulation; (c) Differences between (a) and (b); (d) Comparison of projected 2D $s_{\text{CO}_2}^b$ fields at 30 years.

Fig. 6(d) provides a detailed well-focused analysis through polar projection of the three-dimensional (3D) binary saturation field onto two-dimensional (2D) polar coordinates, illustrated for the final time step (30th year). Each polar plot represents the CO₂ plume extent around the four individual injection wells. This polar projection effectively captures the directional migration patterns and asymmetric plume development characteristic of heterogeneous reservoir conditions. The close alignment between predicted and simulated contours across all four wells demonstrates the model's capability to accurately resolve well-specific plume front geometries, including preferential flow directions influenced by local permeability heterogeneity and structural geological features like faults.

Fig. 7 presents a comprehensive statistical analysis of surrogate model performance across the entire testing dataset, revealing both distributional characteristics and temporal evolution patterns that complement the spatial analysis shown in Fig. 6. In Fig. 7(a), the BCE error distribution exhibits a pronounced right skew with sharp concentration near zero (mean value = 0.099, standard deviation = 0.078), indicating consistent pixel-wise classification accuracy across the dataset. Conversely, in Fig. 7(b) the Dice score distribution demonstrates a strong left skew with modal concentration near unity (mean value = 0.799, standard deviation = 0.181), confirming excellent spatial overlap between the predicted and actual CO₂ plume boundaries. The temporal error analysis reveals contrasting patterns between the two metrics. Fig. 7(c) illustrates systematic BCE error degradation throughout the 30-timestep injection sequence, with median values progressively increasing from approximately 0.02 to 0.18 and expanding interquartile ranges. This trend reflects cumulative uncertainty propagation as CO₂ plume interfaces become increasingly complex during migration. In contrast, Fig. 7(d) demonstrates remarkable Dice score stability, with median values consistently exceeding 0.80 throughout the injection period, thereby confirming the robust geometric fidelity despite evolving subsurface complexity. These complementary metrics provide distinct insights into model performance: BCE captures fine-scale pixel-wise uncertainty that accumulates over time, while Dice maintains geometric accuracy essential for monitoring applications. The consistently low BCE error (0.099 ± 0.078) and high Dice score (0.799 ± 0.181) collectively support the model's viability for real-time geological carbon storage monitoring in risk assessment applications.

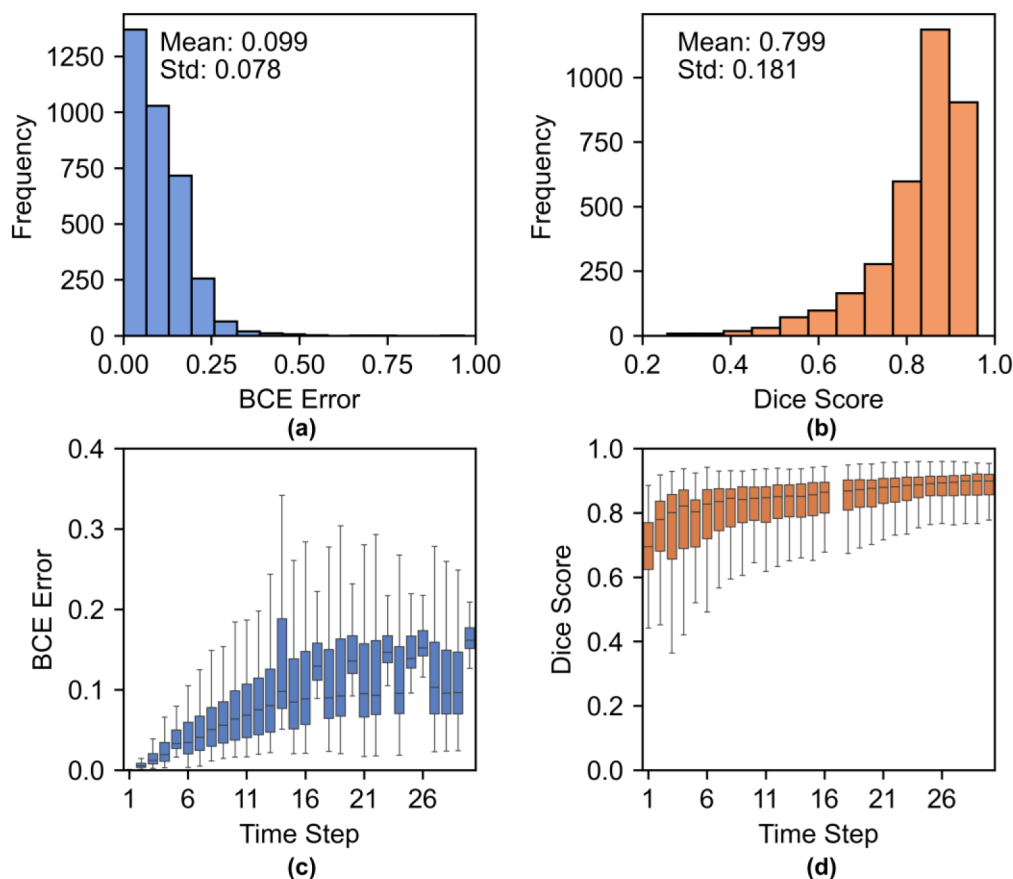


Figure 7—The performance of the surrogate model based on all the testing dataset (a) Distribution of BCE error; (b) Distribution of DICE score; (c) Temporal trend of BCE error; (d) Temporal trend of DICE score.

In terms of computational efficiency, the CNN-Transformer model demonstrates substantial performance gains over conventional simulation methods. When run on GPU (NVIDIA RTX A6000), the model requires only 0.048 seconds per prediction, while the same prediction on CPU (AMD EPYC 7763) takes 0.138 seconds. Both scenarios significantly outperform the conventional reservoir simulator, which requires 479 seconds per simulation on the same CPU platform. This represents speedup factors of approximately 10,000 times (GPU) and 3,500 times (CPU) compared to traditional methods. Notably, even when constrained to CPU-only environments, which may be more representative of typical computational infrastructure, the proposed model maintains exceptional efficiency while delivering comparable accuracy. This computational advantage proves invaluable for large-scale geological carbon storage (GCS) projects that demand extensive what-if scenario evaluation and uncertainty quantification, where thousands of simulation runs may be required. However, it is important to acknowledge that the model's predictive accuracy remains fundamentally constrained by the representativeness and comprehensiveness of the training dataset, which may not fully capture the complex multi-physics processes inherent in GCS operations, including geochemical reactions, geomechanical effects, and their coupled interactions. Future work could address these limitations by incorporating Physics-Informed Neural Networks (PINNs) or similar hybrid components within the framework to explicitly encode additional physical constraints and governing equations, potentially enabling more comprehensive representation of coupled multi-physics phenomena while maintaining computational efficiency.

Therefore, for practical large-scale GCS simulation, we recommend hybrid modeling approaches combining surrogate models for rapid screening with periodic (or low-frequency) high-fidelity simulations for critical decisions, robust uncertainty quantification frameworks, and adaptive training strategies incorporating field monitoring data to maintain model reliability throughout the project lifecycle.

Conclusion

This study presents a novel workflow for efficient CO₂ plume monitoring in geological CO₂ sequestration using a hybrid CNN-Transformer architecture, addressing critical computational demand and practical challenges in large-scale deployment. By integrating the 3D CNN blocks for spatial feature extraction with transformer blocks for sequence modeling, our workflow achieves a unique balance between computational efficiency and physical fidelity essential for operational GCS monitoring.

The key innovation lies in aligning surrogate model outputs with realistic seismicity-detection results through binary saturation mapping at the 10% threshold. This design transforms abstract continuous predictions into actionable intelligence for field operators, while the specialized dual-pathway processing of the CNN-Transformer model eliminates memory-intensive dimensional expansion typical of conventional approaches. Validation was performed based on the complex SPE11-C reservoir model with 258,000 cells (86×50×60 cells), and results demonstrate robust performance with BCE loss of 0.099 ± 0.078 and Dice score of 0.799 ± 0.181 across 116 testing cases, maintaining geometric fidelity throughout 30-year injection periods despite increasing plume complexity.

Our workflow successfully considers critical geological features, including fault-induced migration pathways and heterogeneous flow patterns, providing the spatial resolution necessary for risk assessment in structurally complex reservoirs. The trained surrogate model achieved substantial computational efficiency gains over traditional reservoir simulation, with speedup factors of approximately 10,000 times when deployed on GPU (NVIDIA RTX A6000) and 3,500 times on CPU (AMD EPYC 7763), compared to conventional simulation methods. These performance improvements enable real-time monitoring applications and ensemble-based uncertainty quantification previously infeasible for industrial-scale projects, while maintaining flexibility for deployment across diverse computational infrastructures.

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